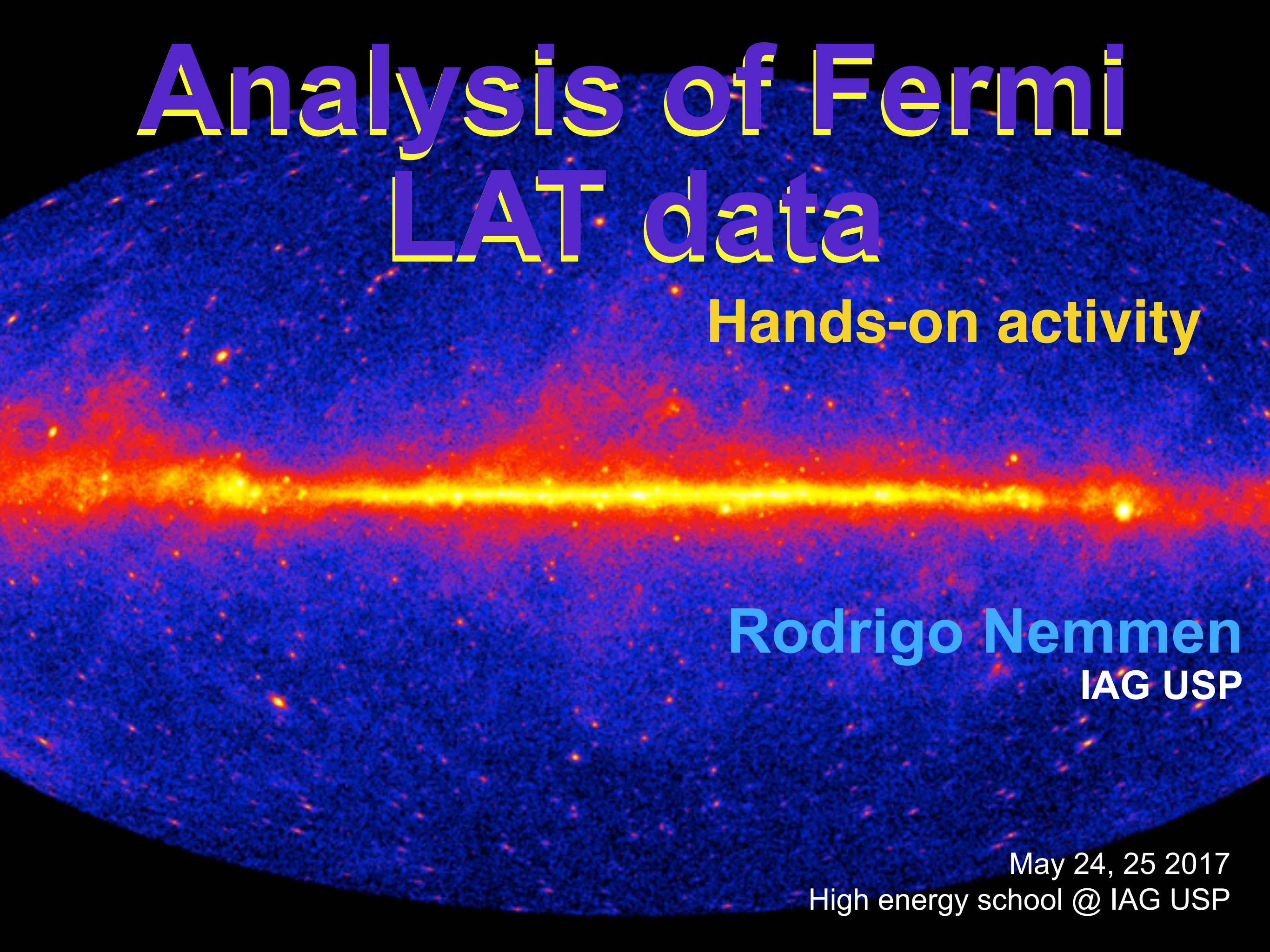


The background of the slide is a Fermi Large Area Telescope (LAT) gamma-ray sky map. It shows a wide field of view of the sky, with a prominent, bright, horizontal band of high-energy gamma-ray emission representing the Milky Way galaxy. The emission is concentrated in the galactic plane and is shown in shades of red, orange, and yellow. The rest of the sky is a deep blue, with numerous small, bright, point-like sources of gamma-ray emission scattered across it. The overall shape of the map is roughly circular, with some distortion at the edges.

Analysis of Fermi LAT data

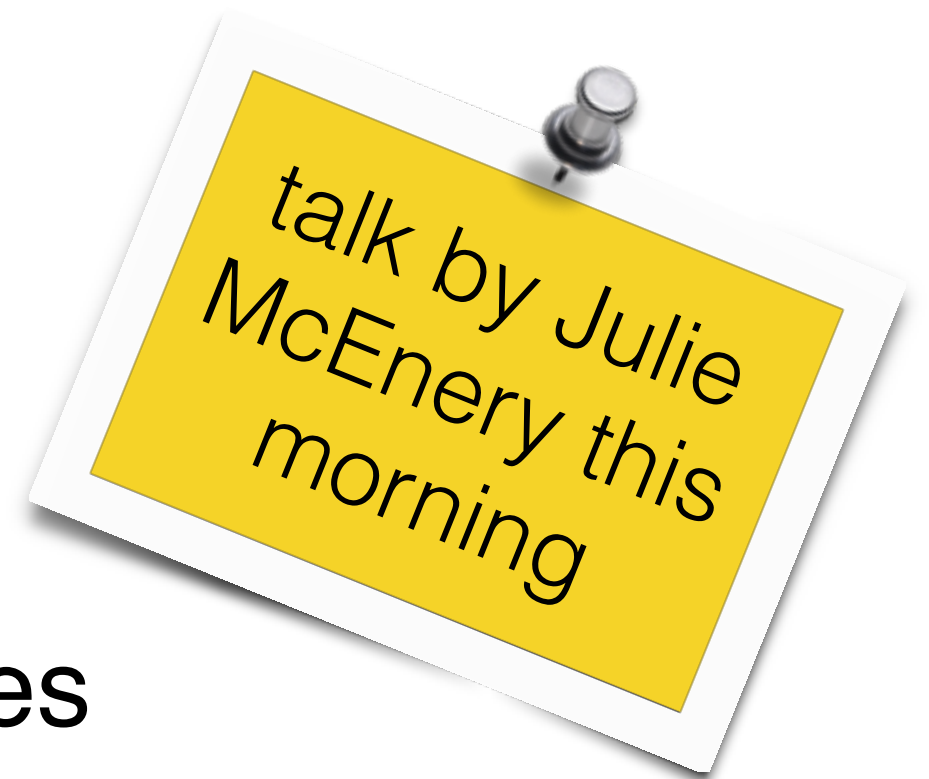
Hands-on activity

Rodrigo Nemmen
IAG USP

May 24, 25 2017
High energy school @ IAG USP

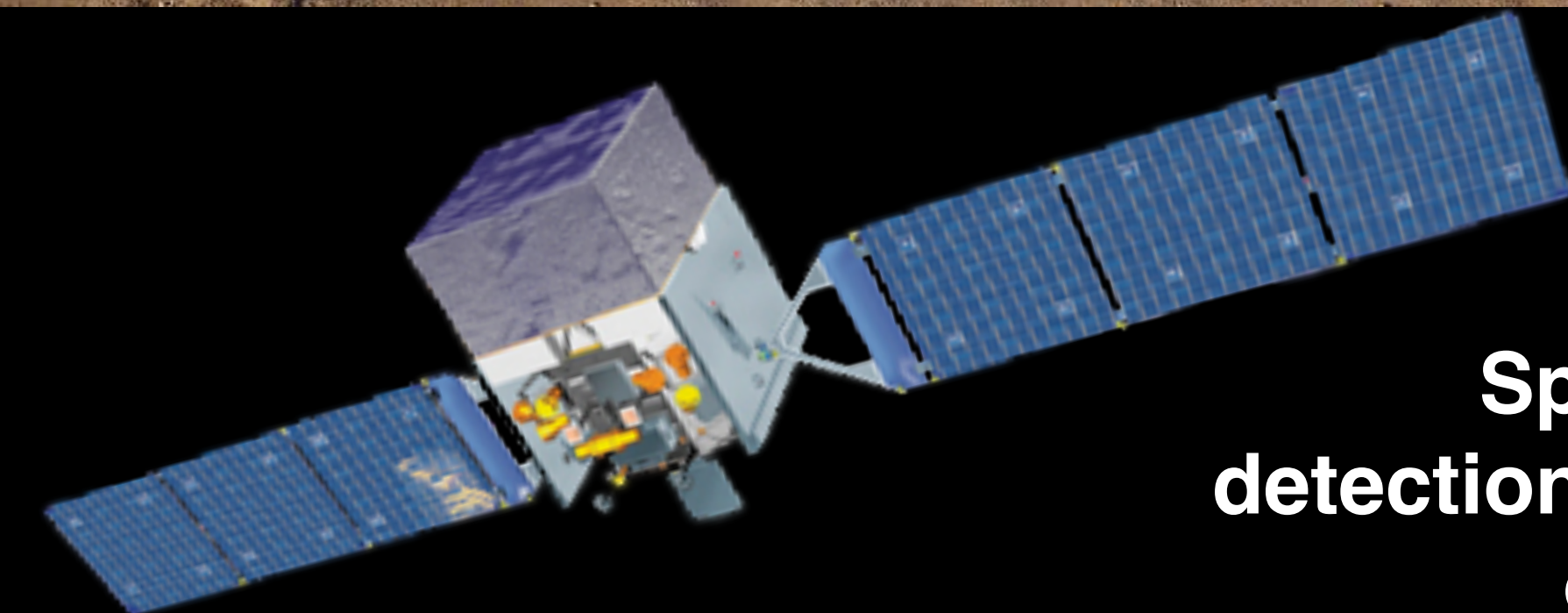
Day 1 activities

- Fermi LAT and how it works
- Fermi LAT science
- Overview of hands-on activities
- Data and software
- **Hands-on: getting started with Fermi LAT data**



Types of gamma-ray observatories

**Ground-based: indirect
detection via Cherenkov
radiation**



**Space-based: direct
detection using techniques
of particle physics**

R. Nemmen, Fermi LAT hands-on

Fermi Gamma Ray Telescope: LSST for high-energy sky, 20 MeV - 300 GeV, whole sky every 3 hours

R. Nemmen

Large Area Telescope (LAT)

Observes 20% of the sky at any instant, entire sky every 3 hrs
20 MeV - 300 GeV

International and interagency collaboration between NASA and DOE in the US and agencies in France, Germany, Italy, Japan and Sweden

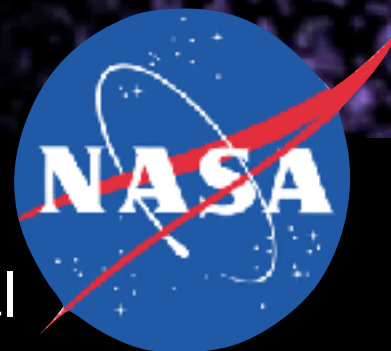
Gamma-ray Burst Monitor (GBM)

Observes entire unocculted sky
Detects transients from 8 keV - 40 MeV

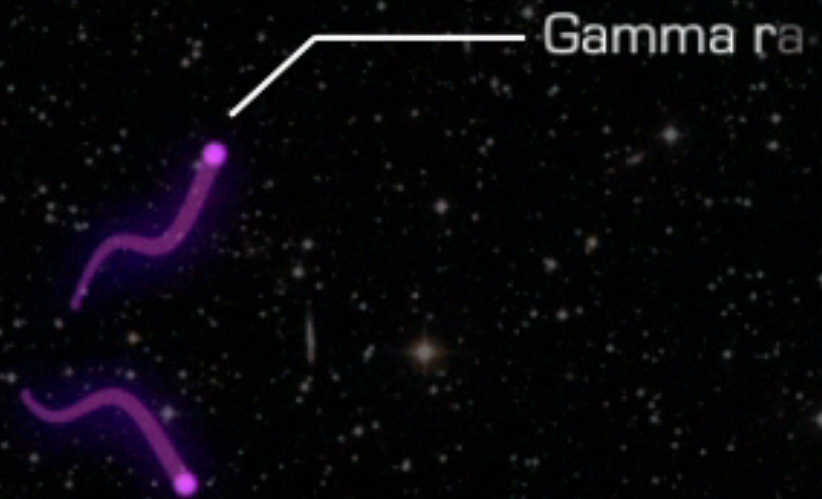
Unique Capabilities for GeV astrophysics

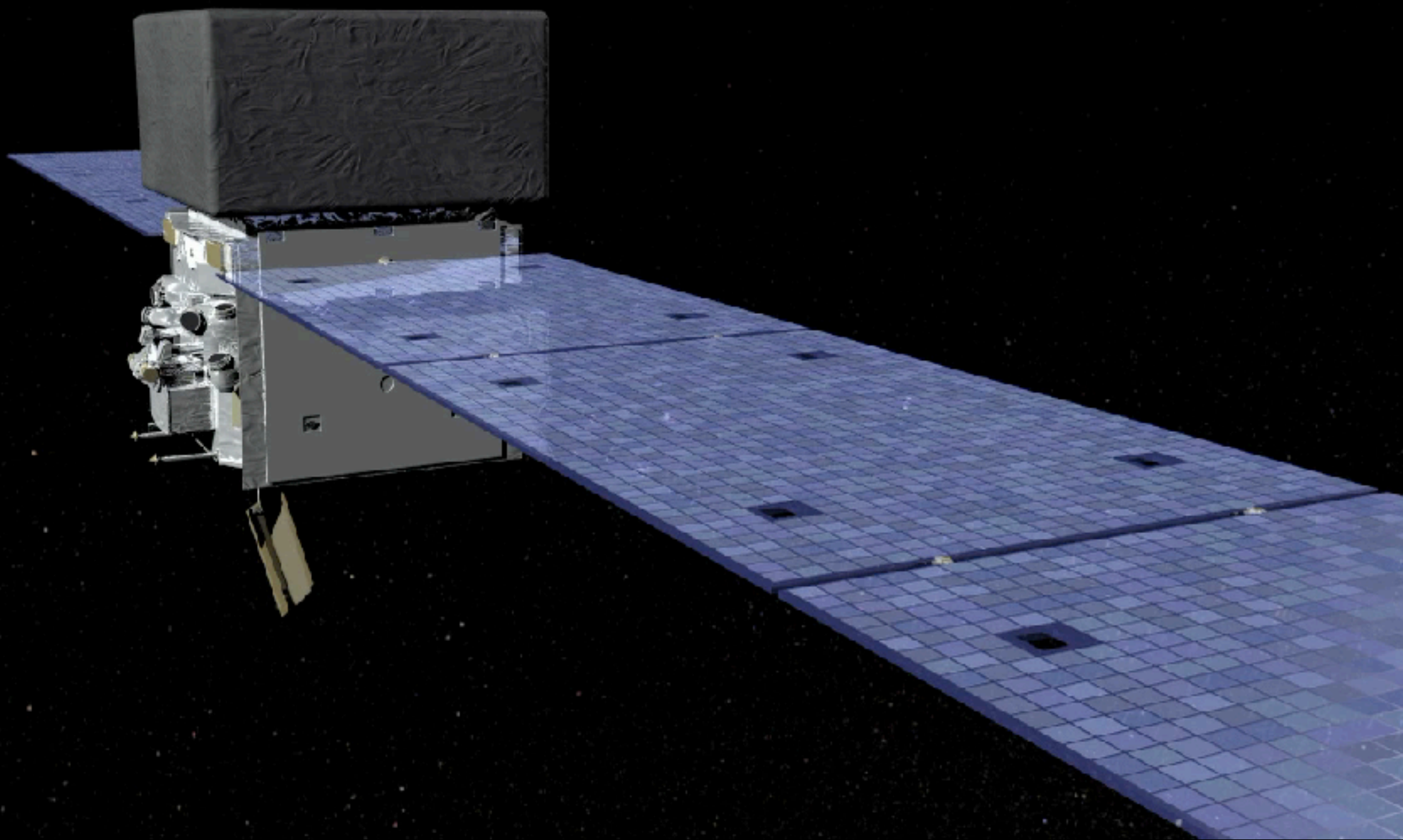
- Large effective area
- Good angular resolution
- Huge energy range
- Wide field of view

Mission Lifetime: 5 year requirement, 10 year goal

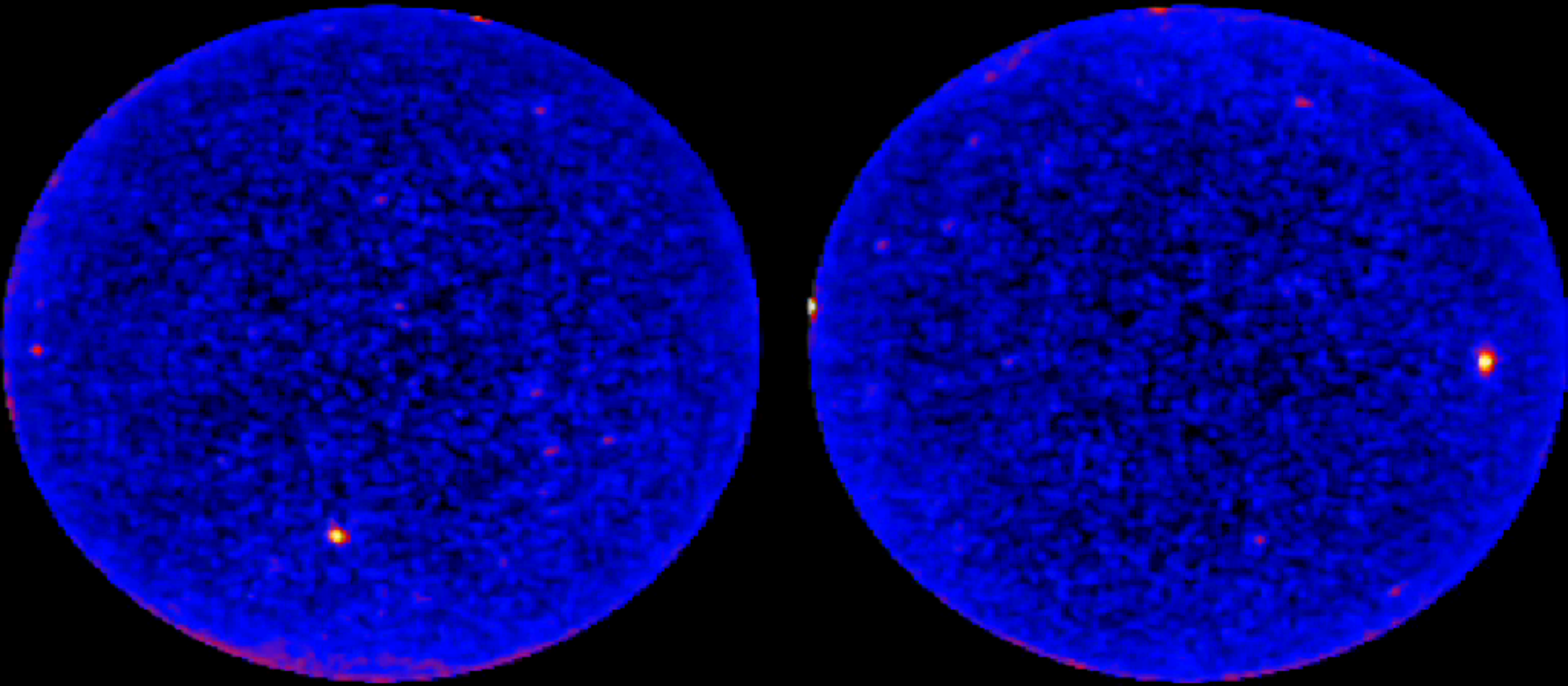


How to Detect a Gamma-ray?

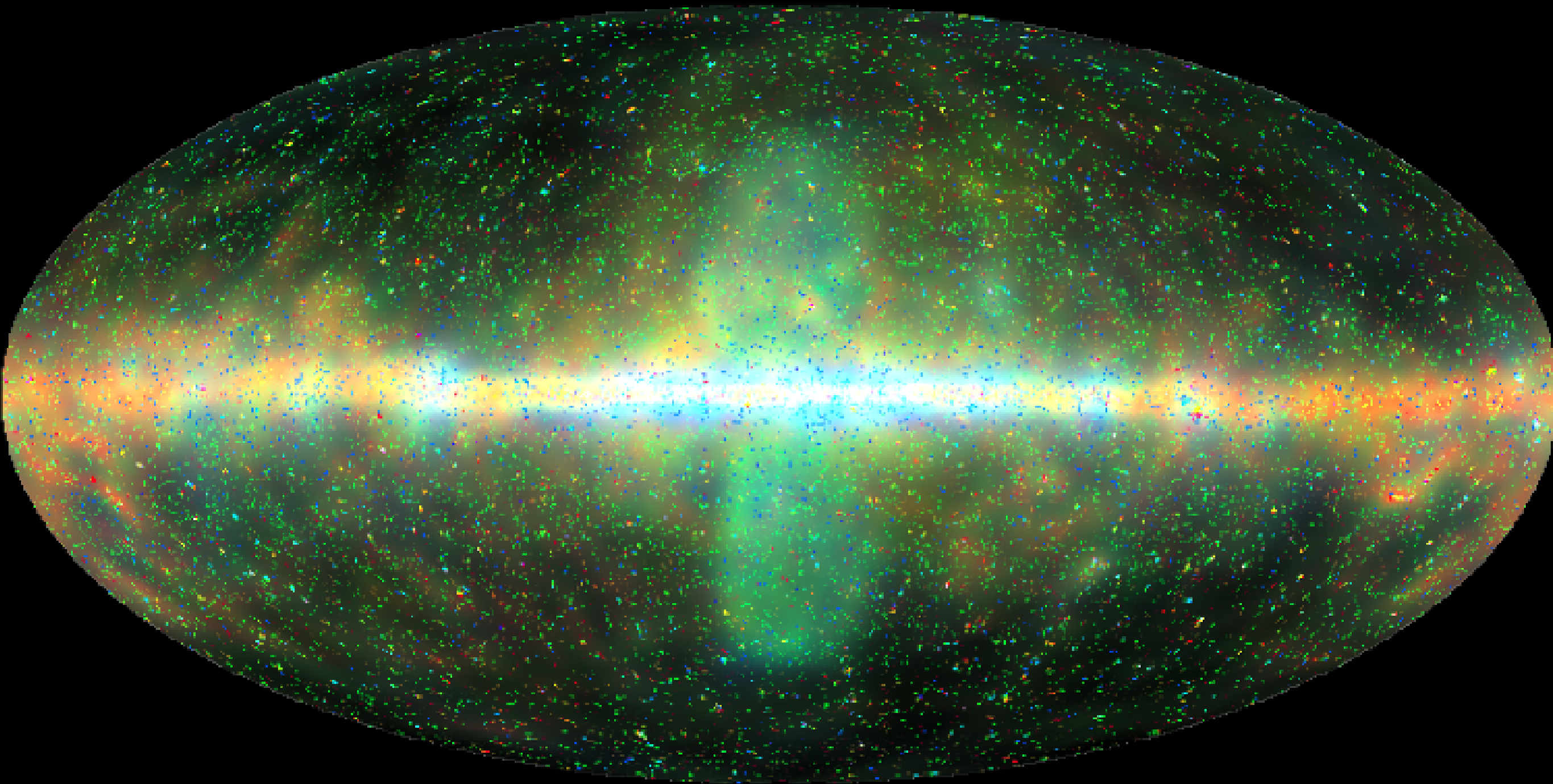




**Fermi LAT: 20 MeV - 300 GeV. Views entire sky
every 3 hours**



Gamma-ray sky after 7 years



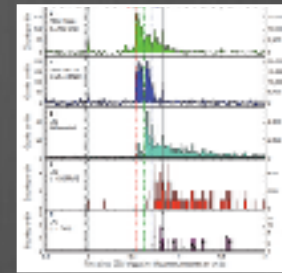
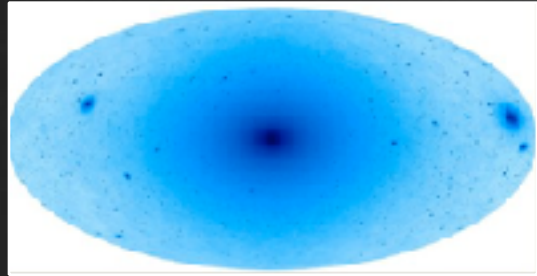
Energies 100 MeV - 300 GeV

Bühler+15

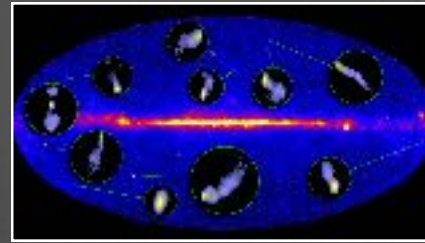
Fermi Highlights and Discoveries

Slide: J. McEnery

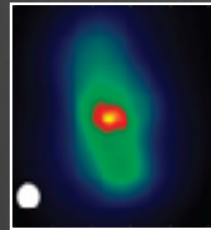
Dark Matter



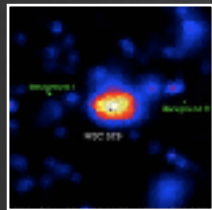
GRBs



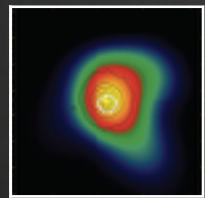
Blazars



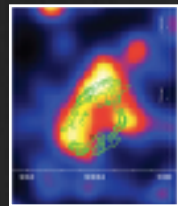
Radio Galaxies



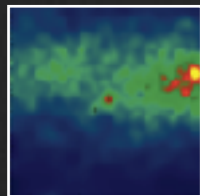
Starburst Galaxies



Globular Clusters



SNRs & PWN



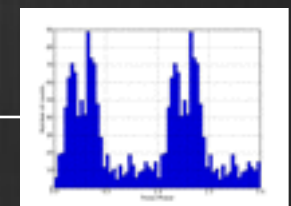
Novae

Galactic

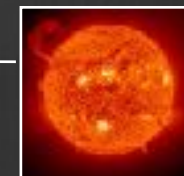
γ -ray Binaries



Pulsars: isolated, binaries, & MSPs



Sun: flares & CR interactions

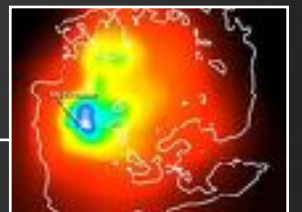


Terrestrial γ -ray Flashes

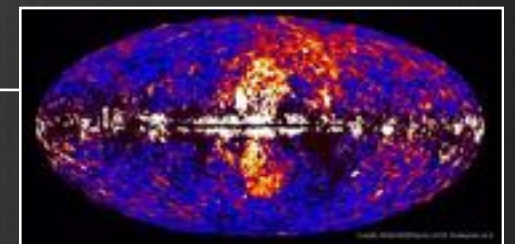


Unidentified Sources

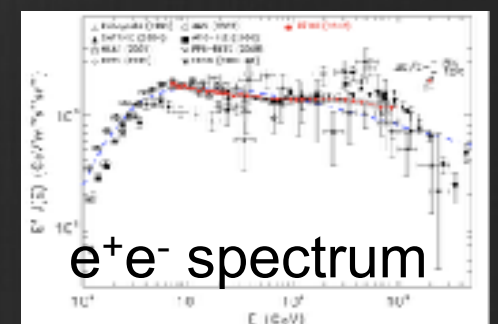
LMC & SMC



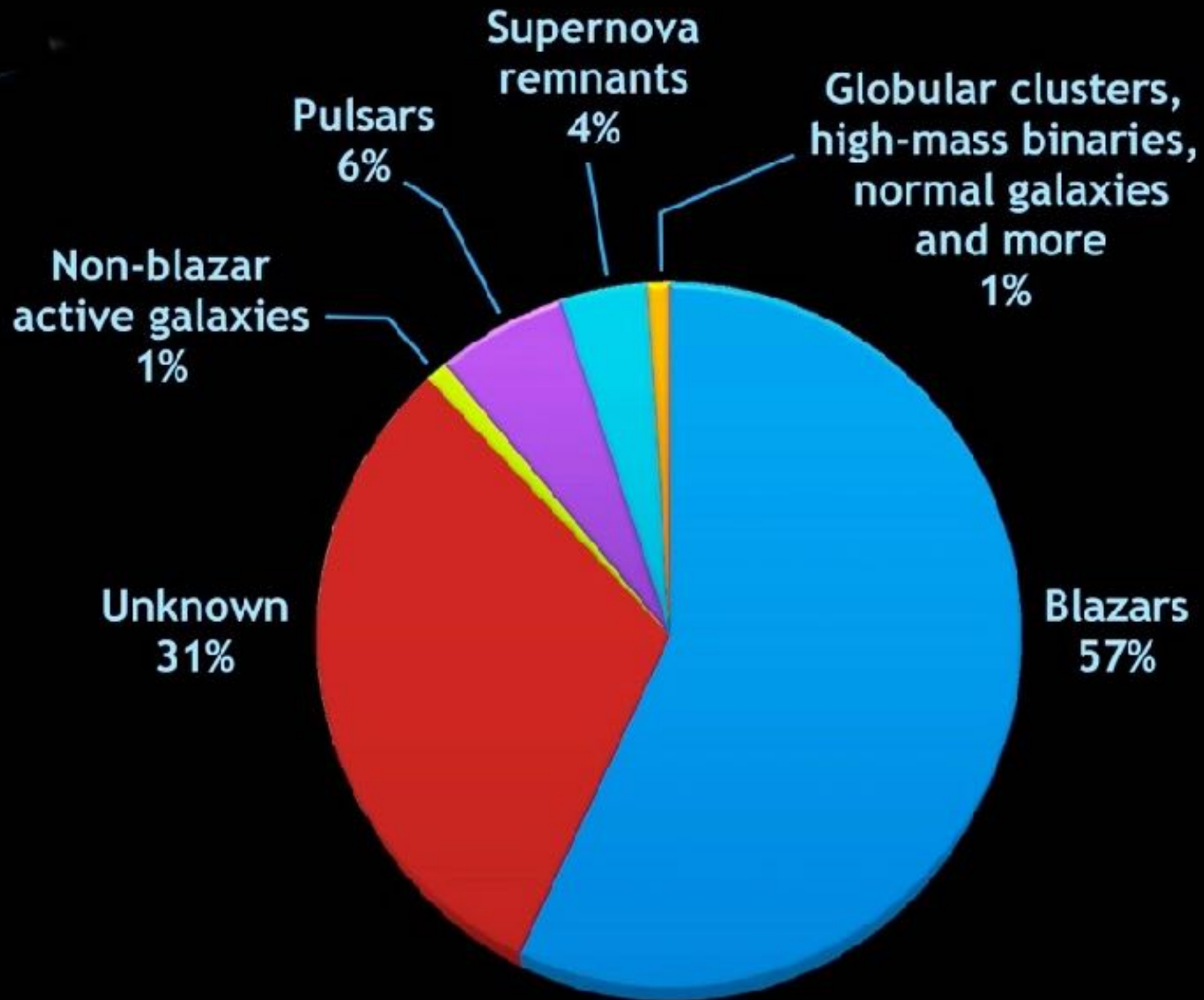
Fermi Bubbles

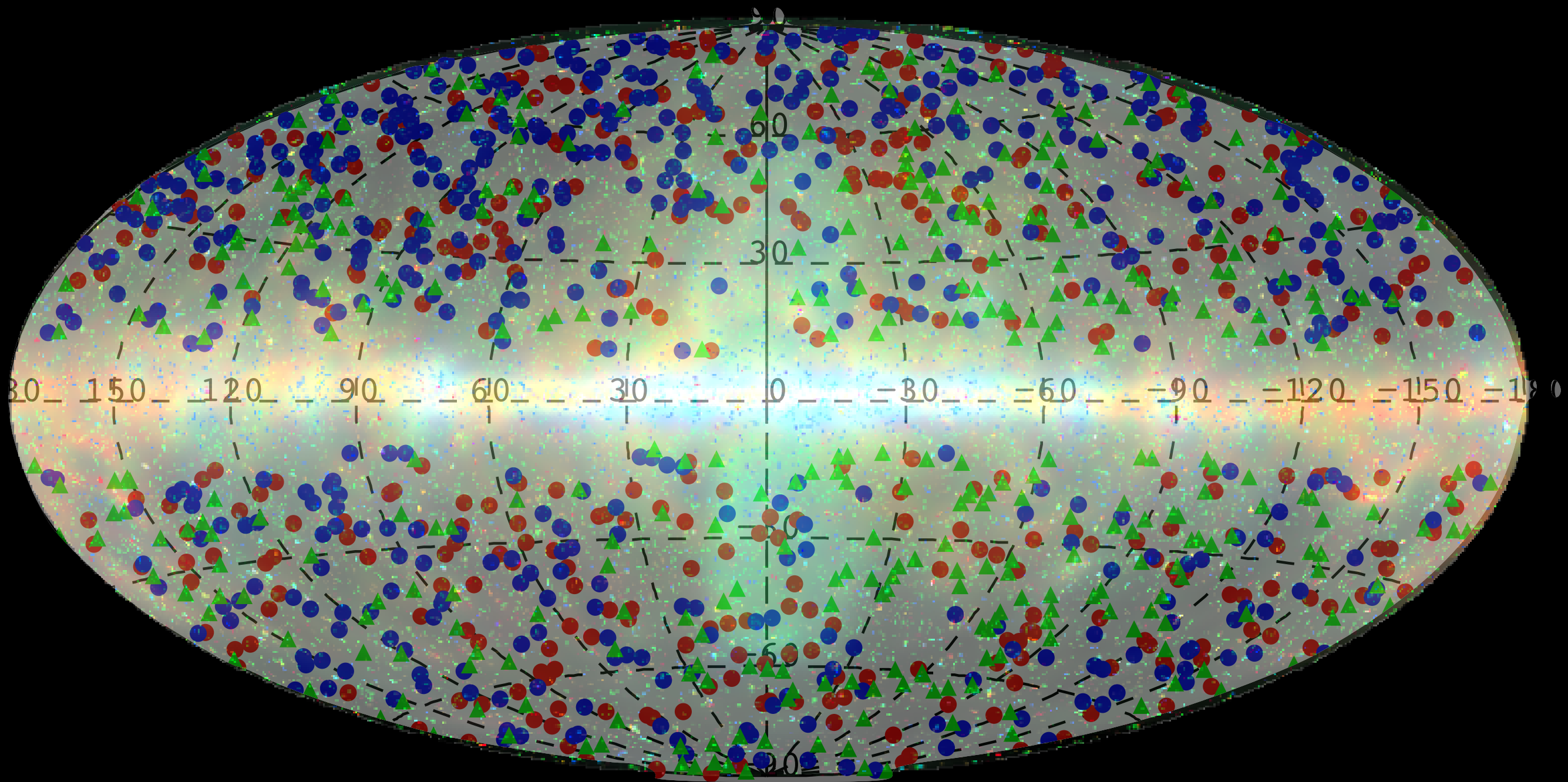


e^+e^- spectrum



Types of sources seen by *Fermi* LAT

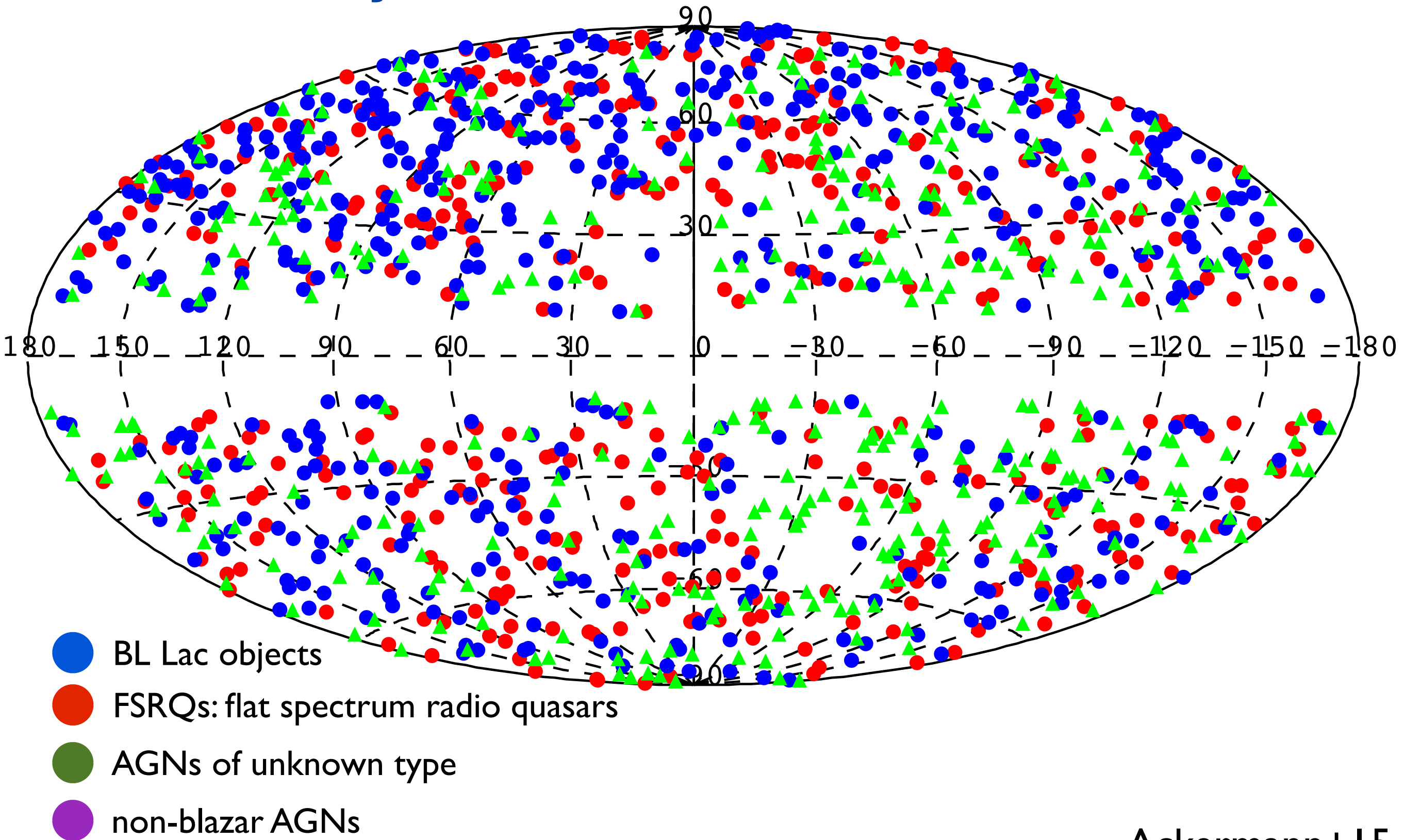




Energies 100 MeV - 300 GeV

Bühler+15

Gamma-ray sky observed by *Fermi* LAT is dominated by blazars



Ackermann+15

Overview of activities

Day 1

- ◆ Introduction, overview
- ◆ Obtaining and exploring LAT data
- ◆ Inspecting the data: counts map

Day 2

- ◆ Introduction: Basics of modeling
- ◆ Likelihood analysis of a blazar
- ◆ Create a SED
- ◆ Produce a light curve

Fermi LAT data

All LAT data products are public

Fermi Science Support Center

<https://fermi.gsfc.nasa.gov/cgi-bin/ssc/LAT/LATDataQuery.cgi>

Data formats:

- FITS files
 - XML source models
 - text-based supporting files
- (more details in the hands-on)

Analysis software tools

- *Fermi ScienceTools*: for general analysis
- *DS9*: FITS file viewer
- *Enrico*: helpful Python scripts for analysis

(more details in the hands-on)

All required software is pre-installed in the VM and ready to use

Tutorial website

<https://github.com/rsnemmen/Fermi-LAT-tutorial>



**Please do not download large files
during the tutorial or the WIFI network
will *overload***

**We will distribute the software and data
you need via USB sticks**





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Bitbucket

nemmen



Blog

<http://astropython.blogspot.com>



figshare

<http://bit.ly/2fax2cT>

Director's cut

(cf separate file)